PDD and Profile of Linear Accelerator

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Contents

1 Objective

1. To determine the percentage depth dose and profile of the Photon and Electron beam from a medical LINAC
2. Write a python code to find various beam parameters from the data obtained during the measurement.

2 Apparatus

- Medical Linear Accelerator
- 3D Scanner (RFA) with Water Reservoir
- Two Ionization Chamber (Field and reference Ionization chamber)
- Electrometer and Connecting cables.
- Thermometer and Barometer
- Leveling tool (Spirit level)

2.1 Medical Linear Accelerator

A medical linear accelerator (linac) is a device that uses high-frequency electromagnetic waves to accelerate charged particles, such as electrons, to high energies. These accelerated particles are then used to produce high-energy X-rays or electron beams for radiation therapy. The linac consists of several components, including an electron gun, a waveguide, and a treatment head. The electron gun generates electrons, which are then accelerated through the waveguide by the electromagnetic waves. The treatment head contains various components, such as a target for X-ray production, collimators to shape the beam, and a multileaf collimator for further beam shaping. The linac is a crucial tool in modern radiation therapy, allowing for precise targeting of tumors while minimizing damage to surrounding healthy tissue.

2.2 3D Scanner (RFA) with Water Reservoir

A 3D Scanner with a water reservoir, often referred to as a Radiation Field Analyzer (RFA), is a device used in radiation therapy to measure and analyze the characteristics of radiation beams. The water reservoir is filled with water, which serves as a medium for simulating human tissue. The RFA is equipped with a scanning mechanism that allows it to move through the water, measuring the dose distribution of the radiation beam at various depths and positions. This information is crucial for treatment planning and quality assurance in radiation therapy, as it helps ensure that the prescribed dose is delivered accurately to the target area while minimizing exposure to surrounding healthy tissues. RFA is a large, usually 50 x 50 x 50cm3 water phantom used to perform radiation dosimetry in a clinical setup. The RFA can hold 166 L of water, and it has a railing on which the chamber can be fixed using holders and moved using a control setup from outside. 3 screws are provided at the base of the RFA to level it. The whole mechanical system is such that a very high accuracy of chamber positioning (0.4 mm) can be achieved (as the ion chambers can move in 3 orthogonal directions).The RFA can provide detailed data on the beam's intensity, shape, and uniformity, which are essential for optimizing treatment outcomes.

2.3 Ionization Chambers

Ionization chambers are devices used to measure ionizing radiation. They consist of a gas-filled chamber with two electrodes. When ionizing radiation passes through the chamber, it ionizes the gas, creating positive ions and free electrons. The electrodes collect these charges, producing an electrical signal that is proportional to the amount of radiation. In radiation therapy, two types of ionization chambers are commonly used: the field ionization chamber and the reference ionization chamber. The field ionization chamber is used to measure the dose distribution of the radiation beam as it scans through the water in the RFA. The reference ionization chamber is typically fixed in a corner of the water tank and monitors the beam output in real-time. The signal from the field ionization chamber is normalized against the reference chamber's signal to remove errors caused by fluctuations in the machine's dose rate, ensuring accurate dose measurements for treatment planning and quality assurance.

2.4 Electrometer and Connecting Cables

An electrometer is a sensitive electronic device used to measure electric charge or current. In the context of radiation therapy, electrometers are used to measure the electrical signals produced by ionization chambers when they detect ionizing radiation. The electrometer amplifies and converts these signals into readable values, which can then be used to calculate the dose of radiation being delivered. Connecting cables are essential for linking the ionization chambers to the electrometer, allowing for the transmission of electrical signals. These cables must be of high quality to ensure accurate signal transmission without interference or loss, which is crucial for obtaining precise measurements in radiation therapy.

2.5 Thermometer and Barometer

A thermometer is an instrument used to measure temperature, while a barometer is used to measure atmospheric pressure. In radiation therapy, both of these instruments are important for ensuring accurate dose measurements. The temperature and pressure of the environment can affect the density of the air in the ionization chambers, which in turn can influence the readings obtained from these chambers. By monitoring and accounting for changes in temperature and pressure, clinicians can apply necessary corrections to the dose measurements, ensuring that the prescribed dose is delivered accurately to the patient. This is particularly important in radiation therapy, where precise dosing is critical for effective treatment and minimizing side effects.

2.6 Leveling Tool (Spirit Level)

A leveling tool, such as a spirit level, is used to ensure that the equipment and setup in radiation therapy are properly aligned and level. Proper leveling is crucial for accurate dose delivery, as any tilt or misalignment can lead to uneven dose distribution and potentially harm healthy tissues while underdosing the target area. The spirit level typically consists of a sealed tube containing a liquid and an air bubble. When the tool is placed on a surface, the position of the bubble indicates whether the surface is level. In radiation therapy, the spirit level can be used to check the alignment of the treatment couch, the water reservoir, and other equipment to ensure that everything is properly positioned for accurate dose delivery. This helps to maintain the precision and effectiveness of the treatment, ultimately improving patient outcomes.

3 Theory

As the beam enters the patient, the absorbed dose at a point varies with depth. This variation depends on many conditions: Beam energy, Field size, distance from the source, and Beam collimation system. Thus, calculating the dose in a patient involves considering these parameters and others as they affect depth dose distribution. Hence, the dose calculation system needs to establish percentage depth dose variation along the central axis of the beam.

The Dose increases with depth as we move into depth from the surface up to a maximum value and then decreases with depth. The dose region between the surface and the depth (d_{max}) of the maximum dose is called the dose build-up region. This increase in dose deposition is due to the increased range of secondary electrons released in the tissue. The larger the photon energy, the larger the range of secondary electrons, and consequently, the larger the depth of dose maximum (d_{max}).

3.1 Beam Parameters

3.1.1 Percentage Depth Dose(PDD)

The Percentage depth dose (PDD) is defined as the quotient, expressed as a percentage, of the absorbed dose at any depth d to the absorbed dose at a reference depth d_0, along the central axis of the beam for a fixed source to surface distance. Mathematically, it can be expressed as:

$$PDD(d) = \frac{D(d)}{D(d_0)} \times 100\% \tag{1}$$

Where $D(d)$ is the absorbed dose at depth d and $D(d_0)$ is the absorbed dose at the reference depth d_0. The reference depth d_0 is typically chosen as the depth of maximum dose (d_{max}) for the given beam energy and field size.

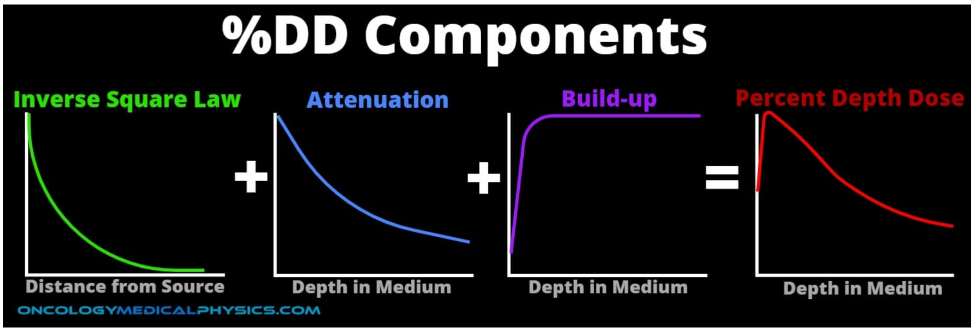


Figure 1: Percentage Depth Dose (PDD) curve showing the variation of absorbed dose with depth for a photon beam. The curve illustrates the build-up region, the depth of maximum dose (d_{max}), and the subsequent decrease in dose with increasing depth due to both inverse square law and attenuation.

3.1.2 Surface Dose

The surface dose is the absorbed dose at the surface of the patient. It is an important parameter in radiation therapy as it can affect the skin and superficial tissues. The surface dose is influenced by factors such as beam energy, field size, and the presence of any bolus material. Generally, for photon beams, the surface dose is lower than the dose at depth due to the build-up effect, where secondary electrons generated by photon interactions deposit their energy at some distance from the point of interaction.

3.1.3 Practical Range

In the case of an electron beam, the depth dose at the end remains constant after a certain depth due to the bremsstrahlung production. So, the quantity 'Practical range' is defined as the depth at which is the depth of the point where the tangent to the descending linear portion of the curve (at the point of inflection) intersects the extrapolated background.

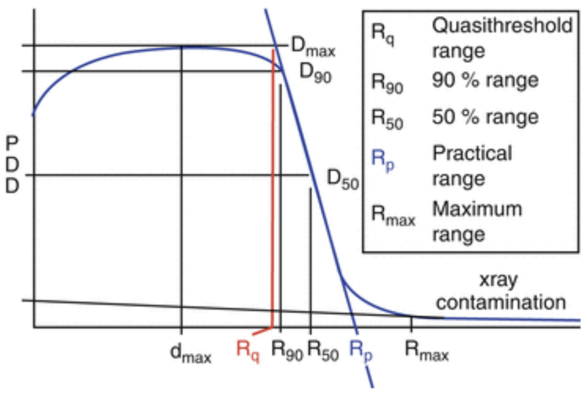


Figure 2: PDD curve for an electron beam showing the practical range, which is the depth at which the dose falls to zero. The curve illustrates the rapid dose fall-off after the practical range due to the limited penetration of electrons in tissue.

3.1.4 Beam Flatness

The beam flatness for the **photon beam** is assessed by finding the maximum D_{max} and minimum D_{min} dose point values on the beam profile within the central 80% of the beam width and then using the following relationship. Standard linac specifications generally require that F be less than 3% when measured in a water phantom at a depth of 10 cm and an SSD of 100 cm for the largest field size available (usually 40 × 40 cm^2).

$$F = \frac{D_{max} - D_{min}}{D_{max} + D_{min}} \times 100\% \tag{2}$$

Flatness for electron beam along major axes at the depth of dose maximum as the separation between 90% and 50% dose points on either side of the beam profile for 10×10 cm^2 applicator and maximum applicator size. The tolerance value is 10 mm.

3.1.5 Beam Symmetry

The beam symmetry for the photon is usually determined at z_{max}, which represents the most sensitive depth for assessment of this beam uniformity parameter. A typical symmetry specification is that any two dose points on a beam profile, equidistant from the central axis point, are within 2% of each other. Alternately, areas under the z_{max} beam profile on each side (left and right) of the central axis extending to the 50% dose level (normalized to 100% at the central axis point) are determined, and symmetry is then calculated from

$$Symmetry = \frac{Area_{left} - Area_{right}}{Area_{left} + Area_{right}} \times 100\% \tag{3}$$

Symmetry for electron beam along major axes at depth of dose maxima in SSD setup (maximum ratio of absorbed doses at symmetrical points from central beam axis and more than 1 cm inside the 90isodose contour).

3.1.6 Penumbra

The photon penumbra is typically defined as the distance between the 80% and 20% dose points on a transverse beam profile measured 10 cm deep in a water phantom. The electron penumbra is usually defined as the distance between the 80% and 20% dose points along a major axis at a given depth. The IEC defines this depth as one-half of the depth of the 80% dose on the central axis.

4 Calculation and Observation

4.1 PDD

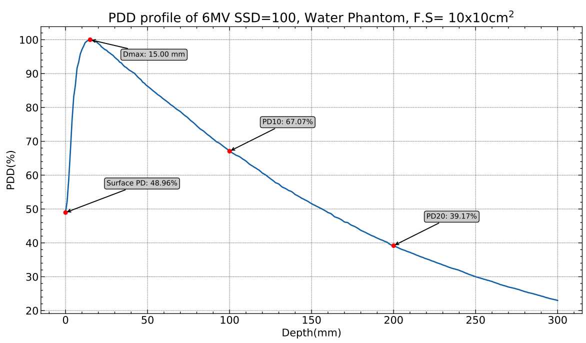
From the experimental data, we can plot the PDD curve for the 6 MV and 10 MV photon beams. The PDD curve illustrates how the absorbed dose varies with depth in a water phantom for a given beam energy and field size. By analyzing the PDD curves, we can determine important parameters such as the surface dose, depth of maximum dose (dmax), and the percentage dose at specific depths (e.g., D10 at 10 cm depth and D20 at 20 cm depth).

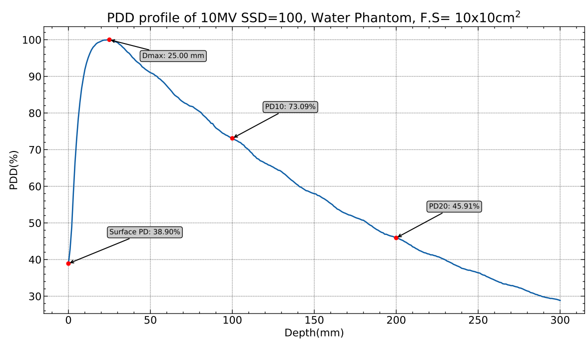
The Data is analysed using Python, and the code snippet for the PDD analysis is provided below. The code reads the depth and dose data from an Excel file, calculates the percentage depth dose, and identifies key parameters such as surface dose, dmax, D10, and D20. Finally, it plots the PDD curve with annotations for these parameters.

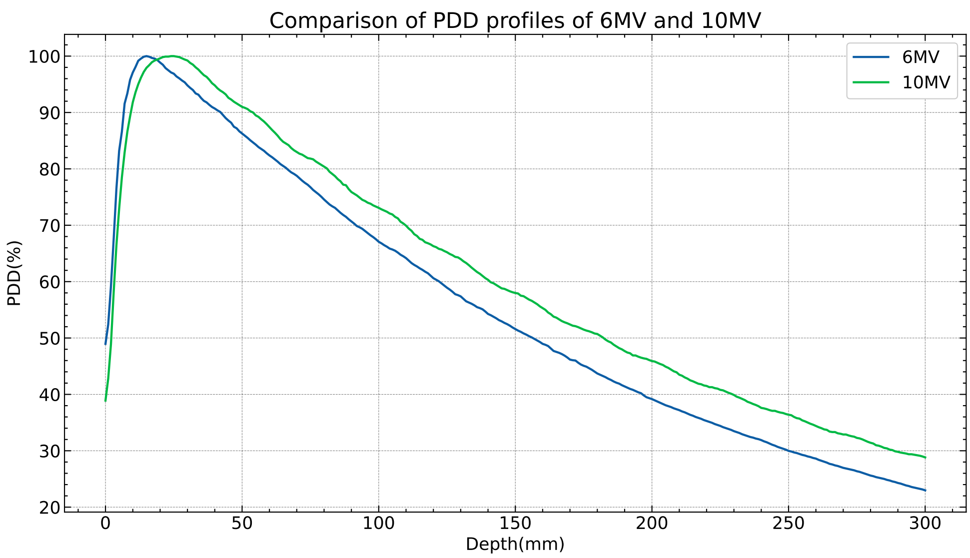
Listing 1: PDD analysis code

```python
# Importing some useful library
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import scienceplots
plt.style.use(["science", "notebook", "grid"])

# Data Extraction from the xlsx file
df = pd.read_excel("PDD and Profile/Program/PDD & PROFILE.xlsx", sheet_name=1)
x = np.array(df["Depth"])
y = np.array(df["Depth dose"])

# Percentage depth dose
PDD = y/max(y)*100

# To find the surface PD(Percentage dose), dmax, D10(Dose at 10 cm depth), D20
# variable initilization
SPD = 0 # surface PD
dmax = 0; D10 = 0;D20 = 0
for i in range(len(PDD)):
    if PDD[i] == 100:
        dmax = x[i]
    if x[i] == 0:
        SPD = PDD[i]
    if x[i] == 100: # 100 mm
```

```
        D10 = PDD[i]
    if x[i] == 200: # 200 mm
        D20 = PDD[i]


#Plotting
plt.figure(figsize=(15, 8))
plt.title(r"PDD profile of 10MV SSD=100, Water Phantom, F.S= 10x10cm$^2$", fontsize=20)
plt.plot(x, PDD)
bbox = dict(boxstyle="round", fc="0.8")
plt.scatter([0, dmax, 100, 200], [SPD, 100, D10, D20], color='red', zorder=3)
plt.annotate(f"Surface PD: {SPD:.2f}%", xy=(0, SPD), xytext=(25, SPD + 8),bbox=bbox,
             arrowprops=dict(arrowstyle="->", lw=1.5), fontsize=11)
plt.annotate(f"Dmax: {dmax:.2f} mm", xy=(dmax, 100), xytext=(dmax + 20, 95),bbox=bbox,
             arrowprops=dict(arrowstyle="->", lw=1.5), fontsize=11)
plt.annotate(f"PD10: {D10:.2f}%", xy=(100, D10), xytext=(120, D10 + 8),bbox=bbox,
             arrowprops=dict(arrowstyle="->", lw=1.5), fontsize=11)
plt.annotate(f"PD20: {D20:.2f}%", xy=(200, D20), xytext=(220, D20 + 8),bbox=bbox,
             arrowprops=dict(arrowstyle="->", lw=1.5), fontsize=11)
plt.ylabel("PDD(%)")
plt.xlabel("Depth(mm)")
plt.savefig("PDD and Profile/Program/PDD_10MV.pdf", dpi=300, bbox_inches='tight')
plt.show()
```







4.1.1 Observation from comparison graph

From the comparison graph of the PDD curves for 6 MV and 10 MV photon beams, we can observe the following:

1. The depth of maximum dose (dmax) is greater for the 10 MV beam compared to the 6 MV beam. This is because higher energy photons have a greater penetration ability, resulting in a deeper dmax.

2. The surface dose is lower for the 10 MV beam compared to the 6 MV beam. This is due to the increased build-up effect for higher energy photons, which causes more dose to be deposited at greater depths rather than at the surface.

3. The percentage depth dose decreases more gradually with depth for the 10 MV beam compared to the 6 MV beam. This is because higher energy photons are less attenuated as they pass through the medium, resulting in a more uniform dose distribution with depth.

4. The D10 and D20 values are higher for the 10 MV beam compared to the 6 MV beam, indicating that a greater percentage of the dose is delivered at deeper depths for the higher energy beam. This is consistent with the increased penetration ability of the 10 MV photons.

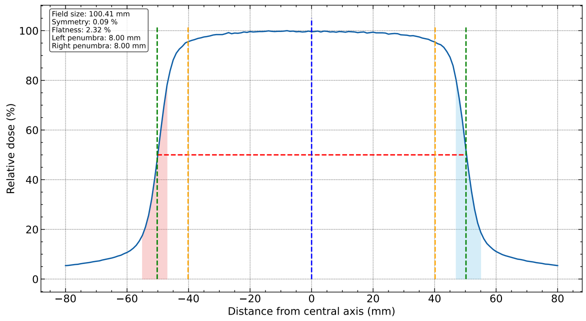
4.2 Beam Profile

The beam profile represents the variation of absorbed dose across the transverse plane of the beam at a specific depth. By analyzing the beam profile, we can assess parameters such as beam flatness and symmetry, which are crucial for ensuring uniform dose distribution in radiation therapy. The code snippet for analyzing the beam profile is provided below. The code reads the transverse dose data from an Excel file, calculates the flatness and symmetry, and plots the beam profile with annotations for these parameters.

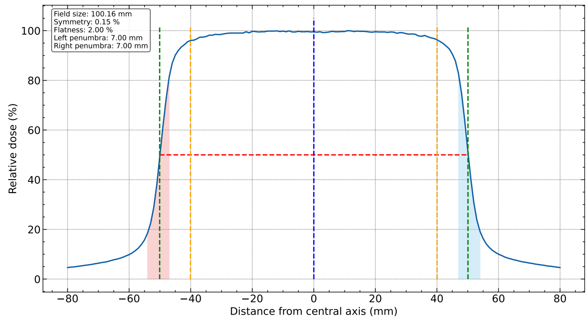
Listing 2: Beam profile analysis code

```python
# Importing some useful library
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from pathlib import Path
import scienceplots
plt.style.use(['science', 'notebook', 'grid'])

# Data Extraction from the xlsx file
base_dir = Path(__file__).resolve().parent
excel_file = base_dir / "PDD & PROFILE.xlsx"
df = pd.read_excel(excel_file, sheet_name=2)
x = np.array(df["Distance from central axis"])
Ix = np.array(df["Crossline dose value"]) #Change the column name (Inline dose value or Crossline dose value)"
Ix=Ix/max(Ix)*100

#Analyzing profile data
xleft = 0; xright = 0    # To find the FWHM(Full width at half maximum) of the inline profile
for i in range(int(len(x)/2)):
    if 55 >= Ix[i] >=45:
        xleft = x[i] - (x[i]-x[i-1])*(Ix[i]-50)/(Ix[i]-Ix[i-1])     #Linear interpolation
    if 55 >= Ix[-i]>=45:
        xright = x[-i] - (x[-i]-x[-i+1])*(Ix[-i]-50)/(Ix[-i]-Ix[-i+1])


FieldWidth = xright - xleft # Geometrical field width


Dmax = 0; Dmin = 100
IIx = []
```

```python
xR = []; leftPenumbra = []; xL = []; rightPenumbra = []
for i in range(len(x)):
    if abs(x[i]) <= 0.4*FieldWidth:
        if Dmax < Ix[i]:
            Dmax = Ix[i]
        if Dmin > Ix[i]:
            Dmin = Ix[i]

    if abs(x[i])<= FieldWidth/2:
        IIx.append(Ix[i])

    if i < int(len(x)/2) and 17 <= Ix[i] <= 83:
        xL.append(x[i])
        leftPenumbra.append(Ix[i])
    if i > int(len(x)/2) and 17 <= Ix[i] <= 83:
        xR.append(x[i])
        rightPenumbra.append(Ix[i])

n = int(len(IIx)/2); h = abs(x[1]-x[0]) # Step size for numerical integration
AreaL = 0.5*h*(IIx[0]+2*sum(IIx[1:n])+IIx[n])           #Trapezoidal rule for area calculation of the left half of the profile
AreaR = 0.5*h*(IIx[n+1]+2*sum(IIx[n+1:-1])+IIx[-1])
perSymmetry = abs((AreaL - AreaR)/(AreaL+AreaR)*100)
perFlatness = abs((Dmax - Dmin)/(Dmax+Dmin)*100)
LeftPenumbra = abs(xL[-1] - xL[0]) # Left penumbra width
RightPenumbra = abs(xR[-1] - xR[0]) # Right penumbra width


#Plotting
fig, ax = plt.subplots(figsize=(15, 8))
ax.plot(x, Ix)
plt.fill_between(xL, leftPenumbra, 0, color='lightcoral', alpha=0.35)
plt.fill_between(xR, rightPenumbra, 0, color='skyblue', alpha=0.35)
plt.plot([-FieldWidth/2, FieldWidth/2], [50, 50], color='red', linestyle='--', label='50% Line')
plt.plot([-FieldWidth/2, -FieldWidth/2], [0, 102], color='green', linestyle='--')
plt.plot([FieldWidth/2, FieldWidth/2], [0, 102], color='green', linestyle='--')
plt.plot([0, 0], [0, 105], color='blue', linestyle='--')
plt.plot([0.4*FieldWidth, 0.4*FieldWidth], [0, 102], color='orange', linestyle='--')
plt.plot([-0.4*FieldWidth, -0.4*FieldWidth], [0, 102], color='orange', linestyle='--')
info_text = (
    f"Field size: {FieldWidth:.2f} mm\n"
    f"Symmetry: {perSymmetry:.2f} %\n"
    f"Flatness: {perFlatness:.2f} %\n"
    f"Left penumbra: {LeftPenumbra:.2f} mm\n"
    f"Right penumbra: {RightPenumbra:.2f} mm"
)
ax.text(0.02, 0.98, info_text, transform=ax.transAxes, va='top', ha='left',
        fontsize=11, bbox=dict(boxstyle='round', facecolor='white', alpha=0.8, edgecolor='black'))
ax.set_xlabel("Distance from central axis (mm)")
ax.set_ylabel("Relative dose (%)")
plt.savefig(base_dir / "CrosslineProfile.pdf", dpi=300, bbox_inches='tight')
plt.show()
```



(a) Crossline beam profile



(b) Inline beam profile

Figure 3: Beam profile plots showing the crossline and inline dose distributions. The distance between the two green vertical lines represents the geometrical field size. The left and right penumbra regions are shown by the reddish and bluish shaded areas, respectively.

5 Conclusion

The PDD profiles for both photon and electron beams were measured and analyzed successfully using the experimental data. The depth dose curves showed the expected variation with depth and allowed the relevant dosimetric parameters to be evaluated. The beam profile analysis was also carried out to determine field size, flatness, symmetry, and penumbra. All the measured beam parameters were found to be within the prescribed tolerance limits. Hence, the photon and electron beams of the linear accelerator were found to be clinically acceptable for treatment use.

References